Computer Networks Lab Report

Submitted By

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Under the guidance of

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of

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in

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FACULTY OF ENGINEERING AND TECHNOLOGY



SRM INSTITUTE OF SCIENCE AND TECHNOLOGY, TIRUCHIRAPPALLI

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Experiment 3: Router Configuration (Creating Passwords, Configuring Interfaces)

Aim

To configure a router by setting console, enable and VTY passwords and by configuring its interfaces with IP addresses, and to verify connectivity between two computers through the router.

Procedure

1. Open Packet Tracer

- Launch Cisco Packet Tracer on your computer.

2. Create a Simple Network

- Drag a router and two computers onto the workspace.
- Connect each computer to the router using Ethernet cables.

3. Access Router CLI

- Click on the router and go to the CLI tab.

4. Set Up Passwords

- Enter global configuration mode:
```
enable
configure terminal
```
- Set the console password, the enable password and the VTY password:
```
line console 0
password cisco
login
enable password cisco
line vty 0 4
password cisco
login
```

5. Configure Router Interfaces

- Enter interface configuration mode, assign an IP address and enable the interface:
```
interface gig0/0
ip address 192.168.1.1 255.255.255.0
no shutdown
```
- Repeat for the second interface:
```
interface gig0/1
ip address 192.168.2.1 255.255.255.0
no shutdown
```

6. Configure IP Addresses on Computers

- Assign 192.168.1.2 and 192.168.2.2 to the first and second computer respectively.

7. Test Connectivity

- Use the ping command to test connectivity between the computers through the router.

Screenshots

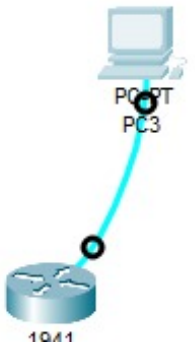


Fig 3.1: Network topology - PC connected to the router

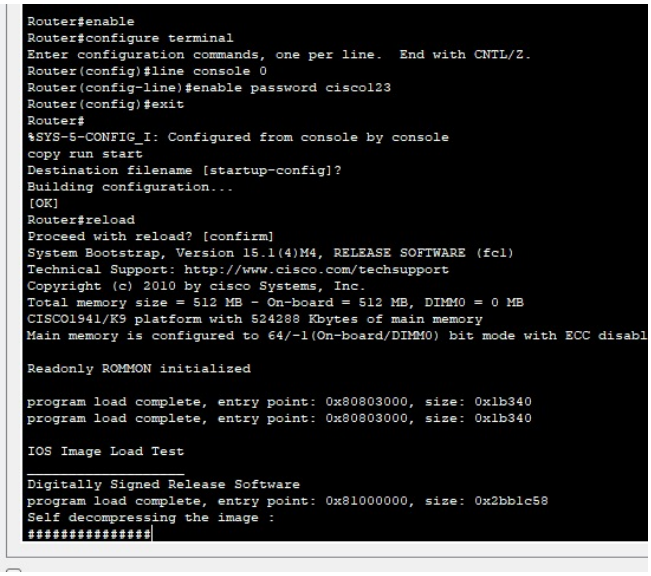


Fig 3.2: Setting the passwords, saving the configuration and reloading the router

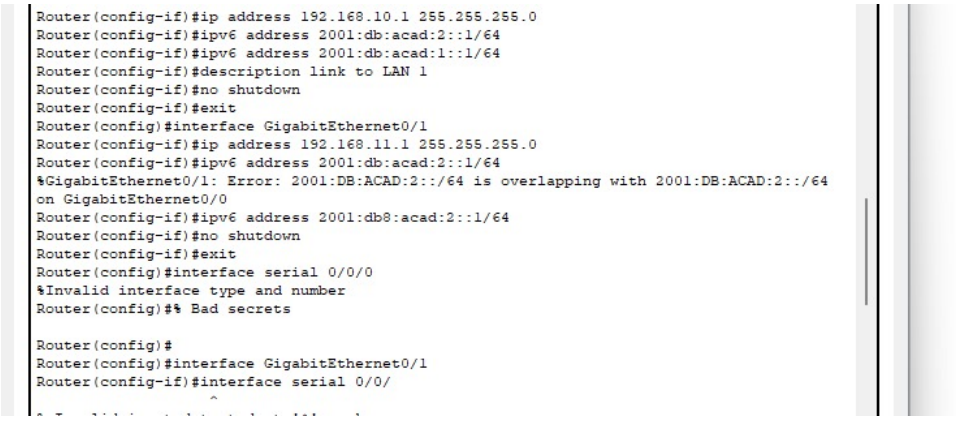


Fig 3.3: Configuring the router interfaces (IPv4 / IPv6 addresses)

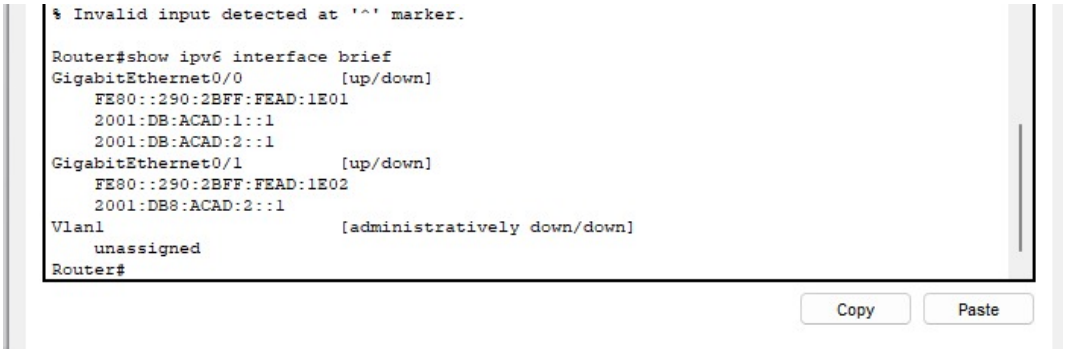


Fig 3.4: Verifying the interface configuration with show ipv6 interface brief

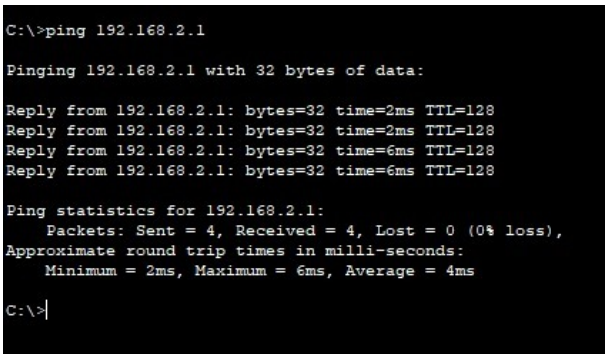


Fig 3.5: Ping from the PC to 192.168.2.1 - 4 packets sent, 4 received, 0% loss

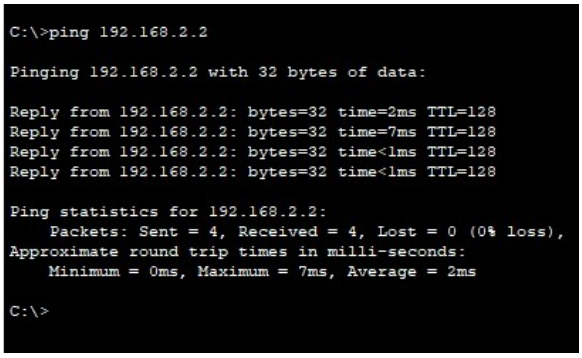


Fig 3.6: Ping to 192.168.2.2 - 4 packets sent, 4 received, 0% loss

Result

The router passwords and interfaces were configured successfully, and the ping test confirmed communication between the computers through the router.

Experiment 4: IP Addressing and Subnetting (VLSM)

Aim

To design a network of three routers connected in a triangle, divide the address block into point-to-point subnets using VLSM, and configure the router interfaces with the calculated addresses.

Procedure

1. Open Packet Tracer

- Launch Cisco Packet Tracer on your computer.

2. Design Network Topology

- Create a network with three routers connected in a triangular fashion.

3. Calculate Subnets Using VLSM

- Determine the number of required subnets and host addresses.
- Divide the network into subnets using VLSM (/30 for each point-to-point link).

4. Assign IP Addresses

- Router 1 to Router 2: 192.168.1.0/30
- Router 1 to Router 3: 192.168.1.4/30
- Router 2 to Router 3: 192.168.1.8/30

5. Configure Interfaces

- Access the CLI of each router.
- Configure the IP address on each interface and enable it:

```
interface <type/number>
ip address <address> 255.255.255.252
no shutdown
```

6. Test Connectivity

- Verify the configuration with show ip interface brief and test connectivity between the routers using ping.

Screenshots

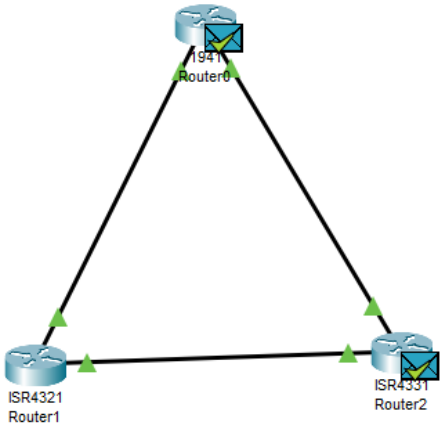


Fig 4.1: Network topology - three routers connected in a triangle

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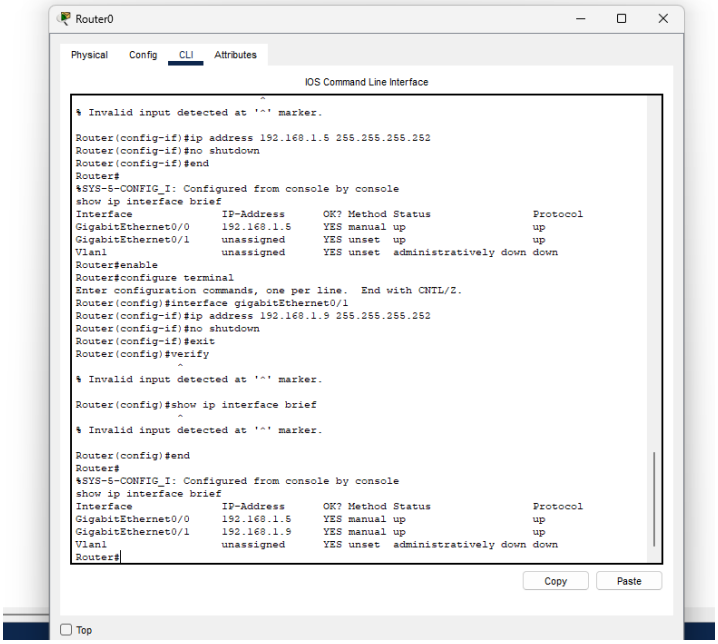


Fig 4.2: Router0 interface configuration and show ip interface brief

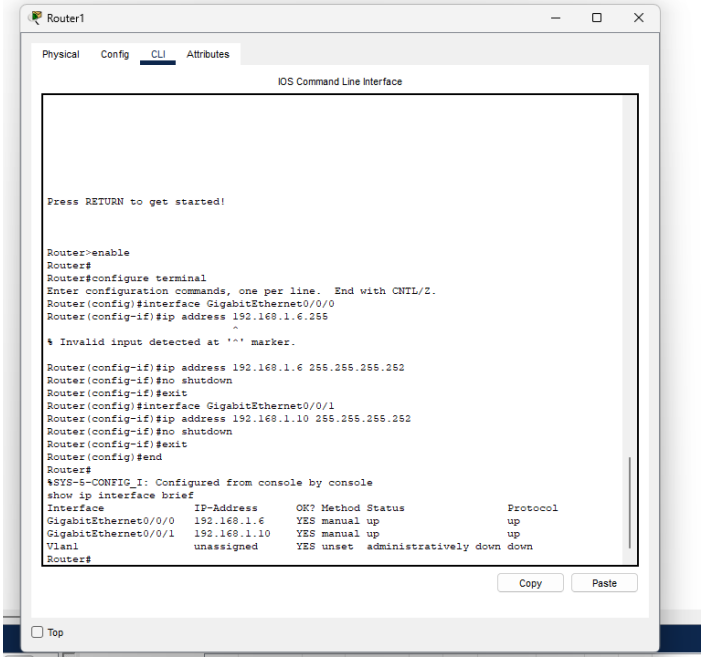


Fig 4.3: Router1 interface configuration and show ip interface brief

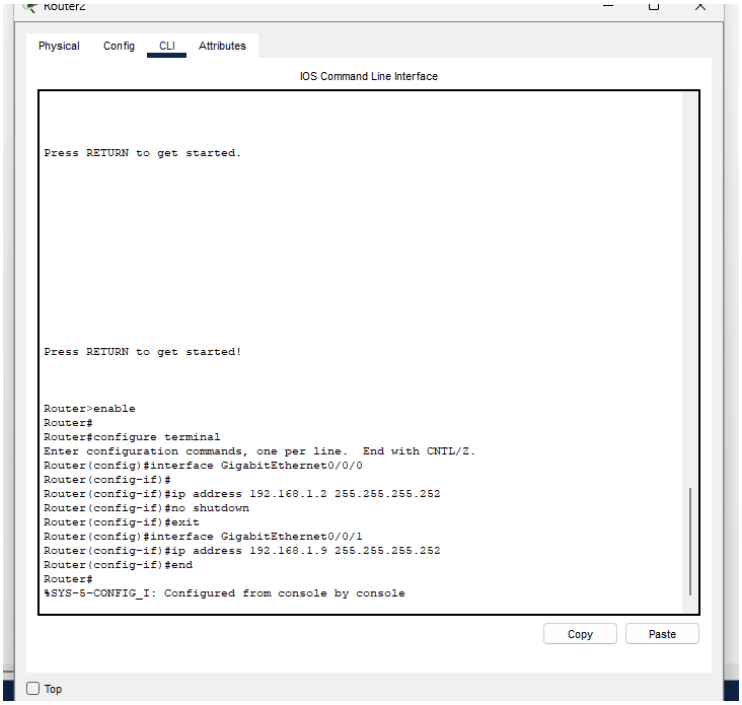


Fig 4.4: Router2 interface configuration

Result

Three routers were connected in a triangular topology, and their interfaces were configured with /30 subnets using VLSM.